

The Problem of Mental Imagery in Language Comprehension

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Introduction

‘The boy threw the dog the bone’; one of the example sentences patient EB was able to understand through the usage of mental imagery (Levine, Calvanio, & Popovics, 1982, cited in p19, Faw, 2009). While the patient was reading the sentence, he formed a visual mental image of the scene. Language comprehension thus seems to involve perceptual qualities; one might be able to reinforce understanding by mentally conjuring up an image. A problem however arises when one believes to have no visual mental imagery at all. Would that entail worse language comprehension?

How exactly one should understand the visual aspects of their mental imagery is a hot debate in philosophical and psychological investigations (see e.g. Brann, 1991). Recently the topic also appeared in public discourse¹. What triggered this public amazement was due to a study by Zeman, Dewar, and Della Sala (2015), where the authors coined the term ‘aphantasia’ for those who lack voluntary visual mental imagery. People from all around the world thus started sharing their inner experiences, claiming to have a blank mind’s eye². This phenomenon raises the reader’s suspicion; would it be the case that when people with alleged ‘aphantasia’ read the sentence ‘The boy threw the dog the bone’, they would not have any visual aspects to their mental imagery? Furthermore, would this also mean that this population has a harder time in language comprehension compared to people with intact visual mental imagery? Such questions will form the foundation of this paper.

This essay acquires ideas from philosophy as well as from psychology. First, two personal interpretations will be lined up on how one might understand mental imagery. These will then be compared to qualitative studies to investigate the first-hand phenomenal experience. Second, quantitative studies from psychology will be enumerated to have a more formal, more scientific, interpretation of the phenomenon. This is then followed by a section on the relationship between mental imagery and language comprehension, paragraphs that lie heavily on the theory of embodied simulation of Bergen (2012), which claims that upon language comprehension one is mentally simulating (or imagining) the contents of a text. The fourth part will then link ‘aphantasia’ to this theory, including experimental findings and possible explanations. Lastly, the conclusion will turn back to some philosophical ideas and will seek to explain ‘aphantasia’ and language comprehension in light of the listed interpretations and evidence.

Phenomenology and the Qualitative Feel

The word ‘phantasia’ roughly means “to appear” or “to make appear...” (p21, Brann, 1991). When one entertains a phantasy, there is something that appears to him in his inner experience. The form of such an experience however differs between individuals. Furthermore, how one is interpreting e.g. the visual aspects of his mental imagery also bares individual differences. Take for example Bill Faw’s personal account on his lack of visual mental imagery (2009). As he put it, he is a “wakeful non-mental-imager” (ibid., p2). Not just he has no visual forms, he also contends not having auditory imagery. On the contrary, spatial (e.g. navigation) and motor imagery seem intact within his phenomenology. Puzzling and complex this may be for those having detailed mental imagery experiences, his perplexity considers the opposite direction: how is it possible that people have visual mental images at all. Taken at its face value, Faw seems to be confident in missing some qualitative experience others possess.

This confidence however is questioned by Schwitzgebel (2002) who argues that when it comes to describing the phenomenology of mental imagery people perform rather badly. That is, Schwitzgebel does not question the experience *per se*, but rather how one thinks and talks about it. To this end, the author gives the reader an imagery task whereby one is asked to

¹ See e.g. the approximately 2600 comments to the following Facebook article:
<https://www.facebook.com/notes/2862324277332876/>

² See e.g. <https://www.facebook.com/groups/204603509580186/>

visualize the main entrance of his home. After this, probing questions follow. The reader is required to dig deep into his own experience, whether one is able to keep in mind two parts of the building at the same time (e.g. a chimney and the door); whether there are focal and peripheral parts of the mental image; whether the peripheral parts have colour or not (and if not, would they be grey e.g.); whether the images fade gradually or abruptly, and so on. As Schwitzgebel argues, most people have troubles answering these questions. As such, confidence in one's mental imagery (or the lack of it) may be unwarranted.

From this it naturally follows to investigate the phenomenology of 'aphantasia'. Watkins for example (2018) describes his own experiences with respect to the blankness of his mind's eyes. Although he claims not having any visual mental imagery, two remarks of his seem meaningful with respect to this essay. First, he contends having "invisible imagery" (ibid., p45), which seems to be resembling to an image although it is not an image, or that it is too faint to confidently describe it as an image. Second, he is on the opinion that the verbal modality in his inner experience is more emphasized than the visual. He admits having auditory experiences although he cannot hear them inside his mind. So, although he is having a certain visual mental image without seeing it and he is also having a certain auditory mental image without hearing it, he still arrives to the conclusion that his verbal (or auditory) experiences are above his visual experiences. Schwitzgebel's scepticism (2002) surfaces again; how mistaken one might be about the phenomenology of one's own mental imagery?

To conclude this section, a qualitative analysis of a case study will be listed. Ganczarek, Żurawska-Żyła, and Rolek (2020) describes A.B. who become gradually aware of her lack of visual mental imagery. A.B. cannot visualize e.g. a horse when asked to. She also finds it a problem not being able to imagine scenes when she is reading a book. According to the study, A.B. lacks both voluntary and involuntary mental imagery. However, "A.B. can only recall blurred contours of objects..." (ibid., p138). These "blurred contours" seem to be resembling to how Watkins described his faint images (2018). According to A.B., the experience she has is just knowing-of-something, rather than picturing-of-something; for example, she knows certain aspects of how her mother looks like, but she cannot "see" these. These descriptions not only showcase how complex mental imagery experiences are, but also directs the attention towards the difficulty of talking about them. When one for example hears the word "horse", how can we be sure that the experience one has is the same (or not) to another's? Is there an objective method that would settle this question? Quantitative research in psychology aims to answer these problems.

Psychological Research

There are two lines of investigations within the quantitative paradigm. Mental imagery is either queried through subjective measures or through objective tasks. As for the first one, the Vividness of Visual Imagery Questionnaire (VVIQ; Marks, 1973, cited in Zeman et al., 2015) is the most common one, which was also used to coin 'aphantasia'. Four different scenes are described to which participants are asked to form a mental image and then rate their experiences on a scale from one to five: one being no mental image whatsoever whereas five being a clear and vivid mental image. As it turns out, Zeman et al.'s participants (2015) scored low on this questionnaire (among other subjective reports). The interpretation is thus followed: there are people who lack voluntary visual mental images.

There are at least two issues rising from this methodology. First, Schwitzgebel (2002) points towards implicit demand characteristics by means of how the instructions are phrased. The VVIQ asks to "think of" a scenario, after which the expression "consider carefully the picture that comes before your mind's eye" (ibid., p44) follows. The questionnaire clearly expects participants to form a visual mental image. Contrary to this, the study of Zeman et al. (2015) originally contacted people with self-ascribed lack of mental imagery. From his introductory email he explicitly formulates his intentions: "We expect you to rate your imagery

at the low end of the spectrum”³. Therefore, although there is one expectation stemming from the questionnaire, a second and contradictory expectation emerged in the original study.

Second, and perhaps more importantly, Schwitzgebel (2002) argues that the VVIQ’s terminology can be interpreted in different ways. What does it mean that a mental image is ‘vivid’? Or that it is ‘clear’? How does one ‘know’ that one only thinks of an object? Most interestingly, Schwitzgebel poses the question: does one understand the expression ‘mind’s eye’ literally or figuratively? Is it the case that there is a (mental) space in front of us that we look at with our mind’s eye, or is it just a metaphor in vain of a better explanation? The current essay’s purpose is not to take a stand on such issues; however, they highlight the problem of various interpretations when it comes to understanding mental imagery.

As for the objective tasks, there are several cognitive tests that can either probe memory for different modalities (e.g. for pictures or words) or more directly testing visual aspects of mental imagery. The most commonly used for the latter is the mental rotation task (Brann, 1991) where participants are shown static images of different shapes and their task is to select the one shape that it matches, albeit rotated to a different position. The reasoning is that when one looks at such a shape and compares it to another one, one mentally rotates it until a match is found. Therefore, visual mental imagery must have been in use.

However, opinions diverge here. The question is how well such objective measures correlate with the aforementioned subjective reports. As per Faw’s findings there is only a “modest” correlation (p61, 2009) between the two. Schwitzgebel is more pessimistic. As he contends, these measures correlate “poorly” or “unsystematically” (p45, 2002). His conclusion is that there is a gap between what these measures show and what people experience phenomenologically.

Theories of Language Comprehension

The paragraphs so far showed that the concept of mental imagery is more problematic than it first seems. Personal accounts on how people experience them diverge; one may claim an absence of visual aspects to it, while another doubts our knowledge over it. Formalizing the study of mental imagery in a quantitative manner is bound to offer clarification, yet it invites further issues. Whether one uses mental imagery in such objective tasks alone is a debate that would require a separate investigation (see e.g. Brann, 1991). Whether one uses mental imagery in language comprehension however is another inquiry and more pressing for the current purposes. After all, Patient EB was able to visually imagine and understand the sentence ‘The boy threw the dog the bone’ (Levine et al., 1982, cited in Faw, 2009). It is yet unknown whether the same applies to populations without any physical or mental disadvantages. The following section investigates this question through the theoretical framework of Bergen’s embodied simulation (2012). Afterwards, the essay turns to ‘aphantasia’ and will seek to find evidence on how people with no mental imagery might process language comprehension.

Bergen (2012) starts by stating that most of our everyday conversations are based on simple things. We use language to describe perceptual qualities; we tell people what happened to us, where it happened, with whom it happened, and so on. Our language can refer to absent things and events. So, for example when one hears ‘The boy threw the dog the bone’, we can have mental concepts of a boy, of a dog, and of a bone, none of which that are present at the verbal utterance of the sentence. Consequently, the listener creates a mental simulation of the description – so claims Bergen’s theory of embodied simulation. This expression ‘mental simulation’ is akin to ‘imagination’. The listener might imagine a boy, a dog, a bone, and the respected action (throwing). Thus, according to Bergen, the listener understands language through this mental simulation. Apparently, there is causation in this claim; first comes the sentence, followed by imagination, resulting in understanding. Strong claims considering the earlier arguments of this essay. Therefore, one requires empirical evidence.

³ See the downloadable supplementary document from the original source, p1.

Bergen lines up several experiments to support his theory. For example, Zwaan, Stanfield, and Yaxley (2002) argue that when people comprehend language, they mentally represent the content differently depending on the shape of an object understood in a sentence. Their participants first read sentences, such as 'eagle in the sky' or 'eagle in a nest'. The authors' reasoning is that when one conceives of an eagle in the sky the animal spreads out his wings, whereas while in his nest the animal is sitting with closed wings. Afterwards, participants were shown pictures where their task was to indicate whether the content of the image was present in the earlier sentence. Crucially, a shape of an image could either match the sentence or not. Results showed that when the shape of the image matched the content of the sentence, people reacted faster than when there was no match. The authors concluded that reading sentences involve mentally creating images of different shapes, and this aids the recognition of consequent pictures.

This piece of evidence is supposed to show that there is a visual element when it comes to understanding language. A sentence however can always have other perceptual modalities. For example, reading 'The boy threw the dog the bone' one might have an auditory experience, imagining how a dog barks. Or just simply reading these sentences might involve inner speech, hearing one's own words. Bergen (2012) refers to this other type of mental (or cognitive) experience as 'verbalizers'. As per the author, one might be a 'visualizer' who memorizes elements of a text by visually imagining them; or a 'verbalizer' who uses auditory experiences to commit something to memory. It can be argued thus that verbalizing is causally connected to understanding as it was the case for visualizing. Perhaps imagining hearing different sounds help one to better comprehend language. As such, Bergen's theory of embodied simulation is not restricted to the visual modality.

Bergen however does not remain biased to his theory. He offers an alternative account, the Mentalese (2012). This is analogous to an actual language that happens in the mind, yet without any reference to e.g. the English language. The difference lies in that it does not have the same visual or auditory form as an actual language. We might think of a dog without using the word form 'dog' or without having an associated auditory experience. Meaning is a key word here; if people use Mentalese the meaning of a 'dog' might mean something different for each and everybody. Crucial to the theory is that individual meanings have a connection to the events in the external world.

Although Bergen is not enthusiastic about this theory (he rather supports the embodied simulation), this idea of a separate meaning linked to an external language is crucial for this essay. Language comprehension rests on an external system, yet our mental experiences representing them are subjective and intimate. As it has already been shown, understanding our own experiences have individual differences. And if someone claims not to have one specific mental experience the others seem to be having (e.g. mental imagery) then how would one go about explaining this? The next section will continue investigating people with 'aphantasia' and their language comprehension.

'Aphantasia' and Language Comprehension

Studies linking 'aphantasia' to language comprehension are scarce since the phenomenon is relatively new. The contemporary focus is rather on verifying how non-imagers differ from imagers in the visual modality. The next best thing one can do is to connect the aforementioned theories to 'aphantasia', looking for empirical evidence closely related to the topic (e.g. memory) and setting up hypotheses and possible predictions.

There is no direct research investigating the theory of embodied simulation (Bergen, 2012) within people with 'aphantasia'. Several hypotheses can be set up however by referring to the previous study (Zwaan et al., 2002). The authors' conclusion was that perceptual elements of a sentence is mentally activated during language comprehension. If, on the one hand, it is the case that people with 'aphantasia' do not have visual (i.e. perceptual) mental imagery, then one hypothesis might be that they do not have mental simulation during language comprehension.

In operational terms, it might be the case that these participants would not respond faster to images that match the preceding sentence compared to non-matching images. If, on the other hand, the problem lies not in 'aphantasia' having no visual mental imagery but rather in the fact that people are mistaken about their inner experiences (as it was argued by Schwitzgebel, 2002), then the original hypothesis might still stand for those who 'have' 'aphantasia' (given that one can still call it as such). That is, they might have similar mental simulations, which would stem from similar reaction times as it was found in the original study. These are just speculations, and the question will have to be settled by future studies.

As Bergen already pointed out (2012) there can be two cognitive styles: visualizers and verbalizers. These ideas have their roots in Pavio's dual coding theory (1990, cited in Monzel, Vetterlein, & Reuter, 2021), according to which memory can either be encoded visually or verbally. As the study's authors indicated, people with 'aphantasia' have impaired visual memory, therefore they investigated whether the same applies to their verbal memory. It was hypothesized that the lack of visual long-term memory might be compensated by verbal long-term memory. Contrary to the author's expectations, 'aphantasic' participants performed worse not just on the visual memory tasks but also on the verbal memory tasks, compared to control participants. This would then suggest that contrary to Pavio's dual coding theory people with 'aphantasia' do not compensate their lack of visual mental imagery by verbal processes. This study however focused on memory processes and not directly on language comprehension. As such it is still unclear whether embodied simulation would be applicable in case of a lack of mental imagery.

Suppose that not having visual mental imagery (nor having verbalization) to aid language comprehension leads to accepting Mentalese (Bergen, 2012). In this case one might argue that an individual language of thought gets activated upon reading or listening. This approach however would not solve the problem of understanding language in 'aphantasia'. The issue in the first place with mental imagery is that it either does not exist (as 'aphantasia' claims) or that it is not understood properly (as Schwitzgebel claims, 2002). As it was shown earlier, claiming that it does not exist evokes methodological issues, both in subjective and objective measures. If then one agrees with Schwitzgebel's arguments (2002), one may claim that the form of mental imagery cannot be defined or described. And this is analogous to the problems with Mentalese; what form does it have? Is it something we are born with (Bergen, 2012)? If so, how does it evolve? If it has no similarity to any languages, how can one define it or talk about it? Thus, one will have to settle down with one of two problematic ideas. Either we will not know the form of mental imagery, or we will not know the form of Mentalese. As Bergen writes (2012) Mentalese is just symbols without any perceptual aspects. One already has a problem in defining perceptual aspects of mental imagery, how would then one go about defining the non-perceptual aspects of a language of thought? Therefore, there seems to be no quick solution to the problem of how mental imagery might be (or not) evoked in language comprehension for those who lack such a mental process.

Conclusion

“Mental imagery is one of the most evident introspective aspects of human thought.” (p1, Palmiero et al., 2019). The current essay’s purpose was to show the opposite; that mental imagery is not that evident, either in introspection or in language comprehension. Faw’s personal account revealed his puzzlement over the possibility that people have mental images while he does not (2009). Schwitzgebel might respond that we are inherently mistaken about our mental imagery experiences (2002). Further qualitative reports showed a possible misconception on what one might experience and how one might talk about it; both Watkins (2018) and Ganczarek et al. (2020) described a sort of blurred or faint mental imagery, a certain experience which is although there, but it still cannot be admitted being image-like. The current essay’s author tends to agree with Schwitzgebel (2002). Upon conjuring up a mental image of my front door, though I have certain visual qualities to my experience (which I cannot describe other than in visual terms), I cannot answer all his questions about it. Nor I can claim with confidence that the experience I have is best labelled as an “image”.

Personal accounts however do not do justice in science. One might try to quantify these experiences via Mark’s VVIQ (1973, cited in Zeman et al., 2015), or take the mental rotation task (Brann, 1991). How one interprets a questionnaire’s instructions or terminology however raises certain concerns. Further to that, Park even claims that there is no need for mental imagery in the mental rotation task (2019). If these measures are not valid and do not correspond to each other – as it was contended by Schwitzgebel (2002) – how can one be sure that when the sentence ‘The boy threw the dog the bone’ is uttered, one does indeed have some sort of visual mental imagery? And to extend the methodological concerns, how would one go about describing the experience of people with ‘aphantasia’?

Two theories were compared in this essay with respect to mental imagery and language comprehension. A topic that has not yet been covered is the distinction between tangible and abstract concepts. The theory of embodied simulation builds on concepts that can be mentally simulated, that is, words having perceptual qualities are easier to imagine compared to abstract words (Bergen, 2012). Bergen analyses metaphors through which one can get a better grasp on how we might simulate (and thus understand) abstract terms. Although the current essay’s scope cannot cover metaphors, it points the attention to an interesting speculation. If people with ‘aphantasia’ are unable to mentally simulate tangible concepts, would this be analogous to how people without ‘aphantasia’ are unable to mentally simulate abstract concepts? I do not claim that the two mental experiences are similar. My best approximation to conceive of how ‘aphantasia’ might not have mental imagery is the following. Perhaps an ‘aphantasic’ hearing the sentence ‘The boy threw the dog the bone’ have a blank mind’s eye similarly to when a ‘non-aphantasic’ hears the sentence “Society demanded justice”. Yet this does not imply that ‘aphantasia’ lacks understanding of tangible concepts, similarly as ‘non-aphantasia’ also does not lack understanding of abstract concepts. Would this then mean that mental imagery (or embodied simulation) is not a necessary part of language comprehension? Furthermore, would this entail an acceptance of Mentalese? Due to its problems mentioned earlier this essay will not move forward with such a conclusion. The goal was only to raise concerns in methodology and in interpretation, not to offer an alternative theory.

One cannot help but wonder whether the problem of mental imagery in language comprehension is really a problem. As Wittgenstein put it “... what value, either for or against mental images, should one attach to these reports?” (p159, Brann, 1991). Since these mental experiences are private, they cannot be compared. The only thing that is publicly accessible is the way we talk about them, therefore Wittgenstein concludes that the only meaningful thing to do is to compare the expressions we use. Perhaps one will never verify whether another person mentally simulates the expression ‘The boy threw the dog the bone’, but at least we can compare our own interpretations.

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